Autonomous UAV Control in Adversary Scenarios: An Application of Reachability Analysis via Controlled Dynamical Embeddings

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***Abstract*— In the reach-avoid scenario in which a defender quad-rotor UAV is preventing an attacking UAV from reaching a critical area, the defender controller must be formulated so that it enforces the defense conditions tractably. NMPC-based controllers are used in scenarios with non-linear dynamics due to the flexibility of the NMPC objective function and trajectory planning. However, defense conditions are typically formulated using reachable sets, which is challenging to reconcile with the NMPC objective, a differentiable scalar function.**

This work proposes a novel NMPC-based defender controller that optimizes the trajectory to best enforce the defense condition derived from the attacker and defender reachable sets. By combining a parametric reachable set framework with the Kullback-Leibler Divergence, the set operations that characterize the defense strategy are expressed as a scalar valued objective function. Additionally, this approach was validated in a simulated environment in MATLAB.

I. Introduction

Reach-Avoid (RA) games are a class of differential games between two adversary agents, wherein an attacker agent aims to reach a target, while the defender agent aims to intercept the attacker before it reaches its goal.

The focus of this work is on three-dimensional reach-avoid games between quad-rotor Unmanned Aerial Vehicles (UAVs), which, importantly, can model real-world scenarios such as UAV incursions into urban airspaces (e.g. airport airspaces [1]). In this scenario, solving RA games primarily implies dealing with non-linear system dynamics and formulating defense strategies that balance efficacy with computational feasibility.

Since the conception of differential games [2], many methods have been developed to solve them with varying tradeoffs and advantages. For instance, solving the Hamilton-Jacobi-Isaacs (HJI) system of differential equations is one method to solve RA games that guarantees optimal control, at the cost of being prohibitively computationally expensive for systems of higher dimensions [3], [4], [5]. Alternatively, Control Barrier Function (CBF) based strategies condition the input as to ensure the system stays within a set of values which ensure interception under certain conditions [4], [6], [7], [8], but the CBF cannot provide an optimal trajectory, only instantaneous inputs, and it struggles to handle non-linear systems.

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Overall, the Non-linear Model Predictive Controller (NMPC) is well-suited for our RA problem since it can deal with non-linear dynamics directly and offers great flexibility in choice of objectives and constraints, by calculating the state trajectory that minimizes an objective function over a finite time horizon [9].

However, there's a mismatch between the normally smooth, differentiable, scalar-valued NMPC objective function and the defense conditions based on sets-of-trajectories traditionally used to formulate defense strategies.

In this light, we propose an NMPC-based defensive control strategy that combines a parametric reachable set framework with the Kullback-Leibler Divergence (KLD) as the NMPC objective function, for a non-linear RA game between UAV quad-rotors. In particular, we make the following contributions:

- A formulation of the RA scenario, in Section II.
- An NMPC based defense strategy which provides a best-effort reach-capture solution, in Section III.
- We introduce the "safe reachability cone" as a tool to validate a defensive trajectory, in Section III-B.
- We validate this control strategy in a MATLAB simulated environment, in Section IV.

Notation: The set of real numbers is denoted by $\mathbb{R}$. The identity matrix is denoted by $\mathbb{I}$. Given a vector $\mathbf{v} \in \mathbb{R}^n$, its transpose is denoted by $\mathbf{v}^T$, and likewise for a matrix $\mathbf{M}$. The inner product between two matrices or vectors is denoted by $\mathbf{v} \cdot \mathbf{M}$. The cross product operation between a vector and a matrix, denoted by $\mathbf{v} \times \mathbf{M}$, is equivalent to the inner product between the skew asymmetric matrix ($SO(3)$) of that vector, which is denoted by $[\mathbf{v}]_\times$, and that matrix. When applied to a matrix, $[\mathbf{M}]_\times$, it can also be used to represent the inverse transformation from an $SO(3)$ matrix to an $\mathbb{R}^3$ vector. The operation vec($\mathbf{M}$) returns the vectorized matrix.

II. Attacker-Defender Problem Formulation

The conditions of the reach-avoid scenario are firstly formulated, starting with the definition of the quad-rotor model in Subsection II-A. This is followed, in Subsection II-B, by the definition of the attacker's goal and the conditions the defender must satisfy.

The scenario in question is a reach-avoid game between a defender drone and an attacker drone. The attacker's goal is to reach a critical area, and it may take any feasible trajectory that accomplishes this goal in reasonable time. The defender's goal is to reach the attacker before it reaches the critical area, despite the exact attacker trajectory not being known ahead of time. The challenge lies in ensuring that the

set of possible defender trajectories fully intersects with the set of attacker trajectories at some moment in the future, thus ensuring the defender reaches the attacker and guaranteeing area defense.

A. Quad-Rotor Model

The six degrees of freedom (6DoF) quad-rotor drone model is built on the following assumptions: its mass remains constant; the method of propulsion are four vertical rotors; attitude control is provided by a torque vector; and altitude control by a scalar thrust input. The model is based on the Euler-Angle-Free model described in [10].

The control input to the quad-rotor drone model can be written as $\mathbf{u} \in \mathbb{R}^4$ where u_1 is the net body force and u_2, u_3, u_4 are the body moments. The position $\mathbf{p} \in \mathbb{R}^3$ dynamics is defined by the equation

$$\dot{\mathbf{p}} = \mathbf{v}, \tag{1}$$

and the velocity $\mathbf{v} \in \mathbb{R}^3$ dynamics by the equation

$$\dot{\mathbf{v}} = \mathbf{g} + \frac{u_1}{m}\mathbf{A} \cdot \mathbf{e}_z. \tag{2}$$

The rotation matrix $\mathbf{A} \in \mathbb{R}^{3\times3}$ dynamics are expressed by

$$\dot{\mathbf{A}} = -\mathbf{w} \times \mathbf{A}, \tag{3}$$

and the angular velocity $\mathbf{w}^B \in \mathbb{R}^3$ dynamics are expressed as

$$\dot{\mathbf{w}} = -\mathbf{J}^{-1} \cdot (\mathbf{w} \times \mathbf{J} \cdot \mathbf{w}) + \mathbf{J}^{-1} \cdot \begin{bmatrix} u_2 \\ u_3 \\ u_4 \end{bmatrix}, \tag{4}$$

where $\mathbf{J} \in \mathbb{R}^{3\times3}$ is the moment of inertia matrix referenced to the center of mass.

The defender drone model is represented by the state vector $\mathbf{x} = [\mathbf{p}; \mathbf{v}; \mathrm{vec}(\mathbf{A}); \mathbf{w}]$, considered to be a point-mass at the drone's center of mass.

B. Defender-Attacker Reachability

The defender is a dynamic system $\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u})$ with a bounded control input $\mathbf{u}$, and the set that is guaranteed to contain all possible states for $\mathbf{u} \in \mathcal{U}$ at a future time k is called the defender reachable set and is denoted by Φ_k. An accurate but overly conservative Φ_k for example, is $\Phi_k = \mathbb{R}^n$. Ideally, Φ_k is as accurate to the system's dynamics as possible, while remaining computationally efficient. The set containing all feasible trajectories from the current time $k = 0$ to the prediction horizon $k = N$ is called the Defender Reachability Cone (DRC), denoted by $\mathcal{D}$, and is considered to be the set of consecutive reachable sets in that time frame, that is, $\mathcal{D} = \{\Phi_k\}, \quad k = [0 : N]$. The Attacker Reachability Cone (ARC) and reachable sets are denoted as $\mathcal{A} = \{\Psi_k\}, \quad k = [0 : N]$. The defender can reach the attacker at any point in the future if the following holds:

$$\exists \Phi_i \in \mathcal{D} : \Phi_i \cap \Psi_i \supseteq \Psi_i, \Psi_i \in \mathcal{A}, 0 \leq i \leq N. \tag{5}$$

Intuitively, this means that if the intersection between the defender and attacker reachable sets ($\Phi_i \cap \Psi_i$) is a superset of the attacker reachable set ($\supseteq \Psi_i$) at some moment in the future (i), then the defender can reach all positions the attacker can reach, as long as Φ is an inner approximation and Ψ is an over-approximation.

C. Parametric Reachable Sets

Traditionally, reachable sets for linear dynamic systems are approximated using Constrained Convex Generators (CCG)s [11], [12]. However, CCGs tend to be computationally expensive and using them with non-linear systems requires linearizations of the system dynamics. Therefore the parametric reachable set via controlled dynamical embeddings is better suited for this work, as it doesn't require linearizing the model.

The parametric reachable set, referred to as a parametope [13], is defined as

$$((\mathring{\mathbf{x}}, \alpha, y))_\varepsilon := \{\mathbf{x} \in \mathbb{R}^n : \varepsilon(\mathring{\mathbf{x}}, (\alpha^T \cdot \alpha)^{-1}) \leq y\}, \tag{6}$$

where $\mathring{\mathbf{x}}$ is the center, α the parameter matrix and y is the squared radius, and $\varepsilon(\mathbf{q}, \mathbf{Q})$ the function for an ellipsoid with center $\mathbf{q}$ and shape matrix $\mathbf{Q}$, given by

$$\varepsilon(\mathbf{q}, \mathbf{Q}) = (\mathbf{x} - \mathbf{q})^T \mathbf{Q}^{-1} (\mathbf{x} - \mathbf{q}). \tag{7}$$

The parametope is propagated via the controlled dynamic embedding system

$$\dot{\mathring{\mathbf{x}}} = f(\mathring{\mathbf{x}}, \mathring{\mathbf{u}}), \quad \dot{\alpha} = U(\mathring{\mathbf{x}}, \alpha, y), \quad \dot{y} = E(\mathring{\mathbf{x}}, \alpha, y), \tag{8}$$

where the parameter dynamics function is defined as

$$U = -\alpha \mathrm{Df}(\mathring{\mathbf{x}}), \tag{9}$$

where Df is the jacobian, and the radius dynamics function is defined as

$$E = c \cdot y, \tag{10}$$

c being an expansion coefficient. In [13], c is computed using the Interval Mixed Jacobian (IMJ) of the system which, in turn, is given by the natural inclusion function matrix of the system's jacobian, obtained through Interval Analysis (IA) [14]. Tools such as [15] can efficiently build the smallest IMJ [16], however implementation of these tools lays outside the scope of this work.

It is noted that in systems with partly decoupled dynamics, like the quad-rotor UAV which has decoupled altitude and attitude dynamics, the dynamic components most sensitive to disturbances will define the bounds of the IMJ, and the actual bounds of the less sensitive components are lost. This is noted with the quad-rotor UAV embeddings system, where the relevant spatial dynamics are overshadowed by the attitude dynamics.

The following scaling method is therefore proposed to compute c, using only the positional dynamics of the drone, and it is given by

$$c(k) = \left(\frac{||\mathbf{p}_{\max}(k) - \mathbf{p}(k)||}{\sqrt{y(k-1)}} - 1 \right) / T_s, \tag{11}$$

where which $\mathbf{p}$ is the translational component of the drone's state, $\mathbf{p}_{\max}$ the furthest possible trajectory in the next time-step, and T_s the sampling period of the simulation.

III. Defense Controller

The defense controller is developed in this section with the defense conditions defined in (5), and the DRC and ARC defined as sets of dynamically evolving parametopes.

Firstly, some notions of ellipsoidal calculus and similarity must be introduced. Consider that the spatial ellipsoid of Φ is denoted by Φ^3 and it is given by the intersection between Φ and the $\mathbb{R}^3$ hyperplane [17], this is

$$\Phi^3 = \Phi \cap \mathbb{R}^3, \tag{12}$$

and the same applies for Ψ^3. Computing the intersection of ellipsoids invariably requires the solution of an optimization problem [18], [19] for each prediction step, at each simulation step, which is computationally untenable.

In lieu of ellipsoid intersection, the KLD is used to quantify ellipsoid similarity. The KLD is traditionally used as a measure of similarity between two gaussian distributions, each described by a positive-definite symmetric covariance matrices, and it is a smooth, always positive, scalar-valued operation that tends to zero when the distributions are identical [20]. Furthermore, the KLD is asymmetric [21]. These characteristics make it a good candidate to quantify similarity between ellipsoids. The KLD from Ω_1 to Ω_2 with centers $\mathbf{q}_1$, $\mathbf{q}_2$ and shape matrices $\mathbf{Q}_1$, $\mathbf{Q}_2$ respectively, and dimensions $n \times n$, has the function notation $\kappa(.,.)$ and is given by

$$\begin{aligned}\kappa(\Omega_1, \Omega_2) =& \frac{1}{2}(\mathtt{tr}(\mathbf{Q}_2^{-1}\mathbf{Q}_1) + (\mathbf{q}_2 - \mathbf{q}_1)^T\mathbf{Q}_2^{-1}(\mathbf{q}_2 - \mathbf{q}_1) \\ & - n + \ln\left(\frac{\det \mathbf{Q}_2}{\det \mathbf{Q}_1}\right)).\end{aligned} \tag{13}$$

The array of pairwise KLD operations between elements of $\mathcal{A}$ and $\mathcal{D}$ is denoted by $K(\mathcal{A}, \mathcal{D})$ such that

$$K(\mathcal{A}, \mathcal{D}) := [\kappa(\Psi_k, \Phi_k)], \quad k = [0 : N]. \tag{14}$$

The `min` operator is not continuous, which makes it unsuitable for optimization problems [22]. The LSE function, denoted by $l(.)$, poses a smooth under-approximation of the `min` operator as seen in [23].

Recalling (5), the defender can reach all positions the attacker can reach at some moment i, if $\Phi_i \cap \Psi_i \supseteq \Psi_i$ and Φ_i is an inner approximation. Since the defender set is defined as a parametope, which is an over-approximation, only the first part of the condition may hold, which does not guarantee interception. However, the larger the volume of $\Phi_i^3 \cap \Psi_i^3$, the more likely the defender is to intercept the attacker.

A. Control Architecture

The controller receives as input the current defender state $\mathbf{x}$ and current DRC $\mathcal{D}$, as well as the current attacker state $\mathbf{z}$, reachable set parameter matrix γ and reachable set radius r.

$\mathcal{A}^3$ is pre-computed in full from $\mathbf{z}$, γ and r before being passed as a parameter to the numerical solver, to ease the computational burden. Computing the full $\mathcal{D}$ trajectory is impractical, however, as the objective function becomes far too complex for the solver to handle. For this reason, an approximated ellipsoid Φ' is defined in the defender's case, given by

$$\Phi' = \{\mathbf{x} \in \mathbb{R}^3 : \varepsilon(\mathbf{p}, \mathbf{W}) \leq 1\}, \tag{15}$$

$$\mathbf{W} = w(\mathbf{A}, y), \tag{16}$$

where $\mathbf{p}$ is the positional component of the defender model, $\mathbf{A}$ is the rotation matrix component of the defender model, y the squared radius computed with the proposed scaling method and $\mathbf{W}$ the shape matrix given by the function

$$w(\mathbf{A}, y) = \mathbf{A} \cdot \left(y \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0.1 \end{bmatrix} \right) \cdot \mathbf{A}^T. \tag{17}$$

The intuition here is that $\mathbf{W}$ is a disk, aligned according to the defender's dynamics, scaled with the most pessimistic radius given by y. The set of all 3-dimensional reduced defender ellipsoids for predicted time-steps is denoted $\mathcal{D}'$, that is $\mathcal{D}' = \{\Phi'_k\}, \quad k = [0 : N]$.

The NMPC objective function will receive as input the $\mathcal{A}^3$ defined previously. It will be defined as a composition of several terms. The minimum KLD terms from Ψ_k^3 to Φ'_k, $k = [0 : N]$ are given by

$$d_{AD} = l(K(\mathcal{A}^3, \mathcal{D}')), \tag{18}$$

and its symmetric counterpart is,

$$d_{DA} = l(K(\mathcal{D}', \mathcal{A}^3)). \tag{19}$$

Assume that both values correspond to the pair of concurrent ellipsoids Φ'_i and Ψ_i^3 closest to each other. In the final stage of approach, when there is overlap between $\mathcal{D}'$ and $\mathcal{A}^3$, in order for the first part of the reach condition defined in (5) to be satisfied, then the inequality $d_{AD} \leq d_{DA}$ must be verified. Thus, the system should evolve such that d_{AD} decreases faster than d_{DA}, which is enforced by minimizing d_{AD} and at the same time maximizing d_{DA} in the objective function.

The relevant minimum KLD is the one between the pair of concurrent ellipsoids which are nearest. This is enforced by introducing the term e in the objective function, which is the minimum of the set of ellipsoid center error norms, denoted as

$$e = l([||\mathbf{p}_k - \mathbf{d}_k||_2]), \quad k = [0 : N], \tag{20}$$

where $\mathbf{d}_k$ is the attacker positional variables.

This all comes together in the optimization problem

$$\begin{aligned} \min_{X,U} \quad & eQ_e + d_{AD}Q_{AD} - d_{DA}Q_{DA} + \sum_{k=1}^{N} ||\mathbf{u}_k||_2^2 R \\ s.t.: \quad & \mathbf{x}_0 = \mathbf{x0} \\ & \mathbf{x}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k), \\ & \mathbf{W}_k = w(\mathbf{A}_k, y_k), \\ & \mathbf{x} \in \mathcal{X} \\ & \mathbf{u} \in \mathcal{U} \end{aligned} \tag{21}$$

where N is the prediction horizon, $f(.,.)$ is the quadrotor's dynamics function, and Q_e, Q_{AD}, Q_{DA} and R are the weight matrices. The optimization variables are $X = [\mathbf{x}_k]$ and $U = [\mathbf{u}_k]$, and the parameters are $\mathbf{x0}$, $\mathbf{y} = [y_k]$ and $\mathcal{A}^3$.

B. Safe Reachability Cone

The Safe Reachability Cone (SRC) is the set of trajectories predicted to be denied from the attacker, and is a useful tool to validate and visualize a defensive trajectory. It is represented as

$$\mathcal{S} = [\Upsilon_k], \quad i \leq k \leq N, \tag{22}$$

where Υ_k is the safe reachability set at time-step k, and i is the time-step the defender is predicted to reach the attacker. Υ is a parametope defined as

$$\Upsilon = ((\mathring{\mathbf{x}}_S, \gamma, r))_{\varepsilon}, \tag{23}$$

propagated via the controlled dynamic embedding system

$$\dot{\mathring{\mathbf{x}}}_S = q(\mathring{\mathbf{z}}) + \mathbf{x}_o, \quad \dot{\gamma} = U(\mathring{\mathbf{z}}, \gamma, r), \quad \dot{r} = E(\mathring{\mathbf{z}}, \gamma, r), \tag{24}$$

where $q(.)$ is the attacker dynamics, $\mathbf{z}$ the attacker state vector, γ the attacker parameter matrix with initial value set at α_i, and r is the attacker squared radius with initial value y_i. $\mathbf{x}_o$ is an offset which aligns the central predicted attacker trajectory with Φ_i. The SRC in a simplified reach-avoid scenario is illustrated in Figure 1.

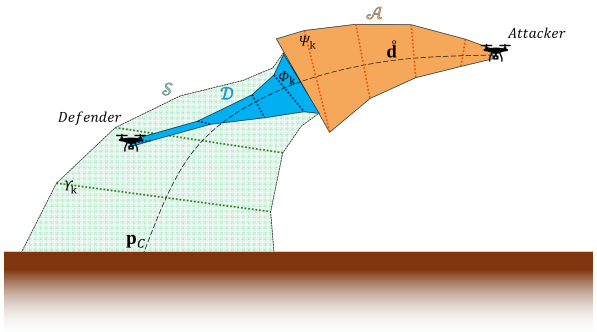


Fig. 1: Illustration of safeguarded area: 2-dimensional simplification. For $k = [0 : N]$, $RD = [\Phi_k]$ is the DRC, $\mathcal{A} = [\Psi_k]$ is the ARC, and $\mathcal{S} = [\Upsilon_k]$ is the SRC.

IV. Simulations and Discussion

The developed systems were validated through simulation, implemented in MATLAB [24], using the Runge-Kutta-4 (RK4) [25] integration method and a time-step length of $0.01s$. The simulated quad-rotor drones have the physical parameters defined in Table I and the input bounds defined in Table II.

A. Reachable Set Validation

A simulation of a quad-rotor model moving from an initial hovering position to another hovering position was run with MATLAB to verify the accuracy of the reachable set, outlined in Section II-C, when compared to potential trajectories. Predicted state trajectories are computed from the current moment to the prediction horizon by fixing the drone's torque input $[u_2; u_3; u_4]$ as combinations of the maximum, minimum, and middle values, and the u_1 input at its minimum or maximum.

TABLE I: Quad-Rotor Parameters

Parameter	Description	Units	Value
M	Total Mass	kg	0.5
L	Arm Length	m	0.25
I_x	Angular Momentum x axis	kg·m^2/s	0.0196
I_y	Angular Momentum y axis	kg·m^2/s	0.0196
I_z	Angular Momentum z axis	kg·m^2/s	0.0264

In both the following figures, the k_{sim} axis indicates the real-time time-steps, and the k_{pred} axis indicates the predicted time-steps. Figure 2 shows the median distance between the reachable set surface and the trajectories it contains [26] at the respective time-steps, where positive values indicate the trajectory is contained in the set, and negative values that it's outside. Figure 3 shows the volume difference between the reachable set and the convex hull formed by the trajectories.

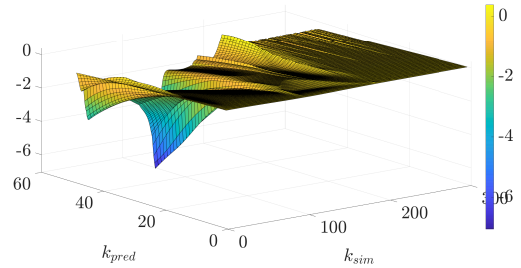


Fig. 2: Median distance between reachable set surface and potential predicted trajectories.

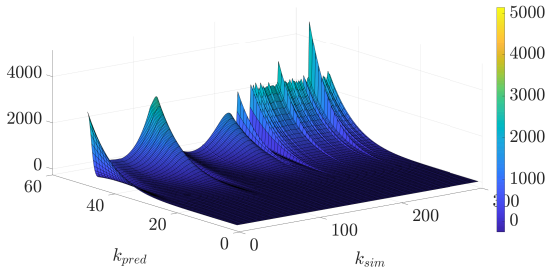


Fig. 3: Volume difference between reachable set and convex hull of trajectories.

B. Two Drone Scenario

A reach-avoid scenario is simulated with MATLAB, in which the attacker drone moves toward the critical area, set at the origin, and has a polynomial trajectory planner in series with a Mellinger controller [27] as the control architecture. Furthermore, the attacker's trajectory is made unpredictable by repeatedly shifting the target waypoint by random amounts. The defender drone is initially hovering above the critical area.

A series of images showing this simulation from the initial moment to the moment the defender reaches the attacker

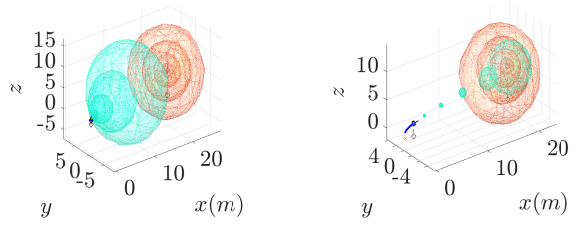
TABLE II: Quad-Rotor Input Bounds

Variable	Units	Lower Bound	Upper Bound
u_1	N	0.0001	50
u_2	rad·s^{-2}	-2π	2π
u_3	rad·s^{-2}	-2π	2π
u_4	rad·s^{-2}	-2π	2π

are in Figure 4, where $\mathcal{D}^3$ are the blue ellipsoids and $\mathcal{A}^3$ are the orange ellipsoids. The simulation shows the defender speeding towards the attacker, then as $\mathcal{D}^3$ and $\mathcal{A}^3$ overlap the defender slows down as there is a Φ_i^3 which expands to contain the corresponding Ψ_i^3. This can be seen as the defender positioning itself such that there is a moment it can reach any trajectory the attacker may take.

The defender's performance data is shown in Figure 5, where the top graph shows the norm of the distance between defender and attacker throughout the simulation, and the bottom graph shows the area of $\mathcal{A}^3 \cap \mathcal{G}$ in blue, and $\mathcal{S}^3 \cap \mathcal{G}$ in orange, denoted by Θ_A and Θ_S respectively, with $\mathcal{G}$ being the $\mathbb{R}^3$ ground plane. It can be seen that Θ_S fully overlaps Θ_A from approximately $t = 0.1s$ to $t = 0.3s$, which indicates that throughout this timeframe the defender is positioned such that it can reach all attacker trajectories.

The evolution of the KLD between the 3-dimensional ARC and DRC is shown in Figure 6 as a surface, where the columns are the $K(\mathcal{A}^3, \mathcal{D}')$ at each simulation time-step, the values being indicated by the color bar, capped at 1000. The red line highlights the minimum KLD at each k_{pred}. It can be seen that the KLD tends to decrease to zero, indicating that the DRC is overlapping with the ARC. Observing red line, it can be seen that the k_{pred} of the minimum KLD tends to decrease, showing how the predicted time of interception approaches the present.



(a) $t = 0.01s$ (b) $t = 0.27s$

(c) $t = 0.54s$ (d) $t = 0.81s$

Fig. 4: Visualization of single defender with defense-centric controller vs unpredictable attacker with Mellinger controller in defense scenario, N=50.

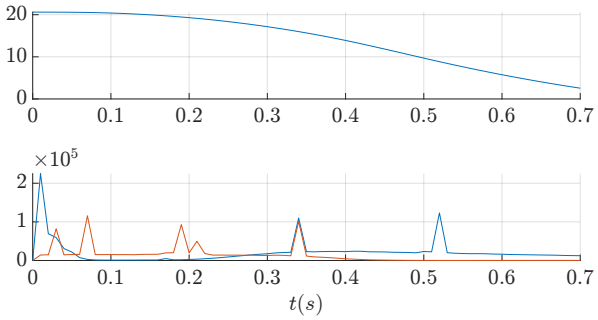


Fig. 5: Performance analysis of defense-centric controller in defense scenario with 1 defender and 1 unpredictable attacker.

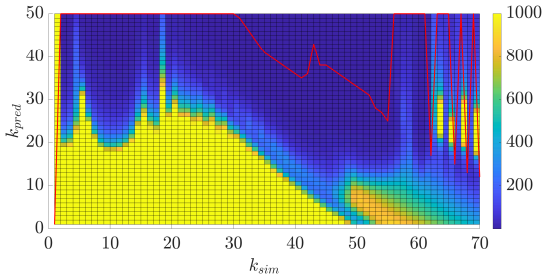


Fig. 6: KLD between A and D across simulation.

V. Conclusions and Further Work

This work tackles the issue of non-linear quad-rotor UAV control in a reach-avoid game. The simulation results in Section IV show that the proposed control strategy correlates with the defender continuously positioning itself such that there is a moment it can reach any trajectory the attacker may take.

In future works, the parametric reachable set propagation could be refined, by using more sophisticated interval arithmetic tools for example, to improve the accuracy of the reachable sets. Furthermore, physical experiments could be conducted to validate the proposed controller in the real world. A tangential, but perhaps more interesting line of research would be to apply the reachability-based control philosophy here proposed to more constructive scenarios, such as search-and-rescue out at sea.

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